

KABILESH S

Electronics & Communication Engineering Student | Embedded Systems Engineer

ECE student (Panimalar Engineering College, 2024–28) with hands-on experience in embedded systems, PCB design, and hardware troubleshooting. Skilled in Arduino, ESP32, UART/SPI/I2C protocols, MATLAB/Simulink, and KiCad. Completed internship at Salcomp Manufacturing India – a leading mobile charger production facility. Seeking core/manufacturing roles as an embedded systems engineer.

EDUCATION

B.E. Electronics & Communication Engineering | 2024 – 2028
Panimalar Engineering College, Chennai

Higher Secondary Education (12th) | 2024
Bharathidasan Mat. Hr. Sec. School, Tamil Nadu

PROJECTS

IoT Battery Monitoring & Digital Twin System

- Built ESP32-based real-time battery monitoring with cloud logging and custom PCB, integrating 4+ sensors — improved data capture accuracy by ~20%.
- Developed AI-driven Digital Twin (MATLAB/Simulink) for SoC, SoH & DTE prediction using hybrid physics + ML model, reducing prediction error by 15%.

IoT Solar Energy Monitoring & Theft Detection

- Designed dual-node IoT system to monitor energy usage and detect power theft using intelligent sensor analysis, achieving real-time fault identification.
- Implemented cloud-based fault detection and smart relay control, reducing system recovery time by ~30%.

Smart AI Ambulance Route Optimization

- ESP32 + GPS system for real-time ambulance tracking and dynamic traffic signal control, reducing average response time by ~25%.
- Built live dashboard for route optimization and emergency corridor management with multi-node communication.

LIFEBAR – Hybrid Mobile-Satellite Communication Device

- Designed portable system integrating GSM/4G and satellite communication for remote connectivity, SOS alerts, and automatic network switching.

EXPERIENCE

Engineering Intern | Aug 2025

Salcomp Manufacturing India Pvt. Ltd., Sriperumbudur

- Assisted in mobile charger manufacturing — PCB assembly, soldering, and quality inspection on production lines.
- Gained exposure to large-scale electronics production, testing procedures, and component integration.
- Supported troubleshooting and validation of electronic modules; understood industrial safety & workflow standards.

Assistant Technician | 2025 – Present

Computer & Laptop Service Center, Kanchipuram

- Diagnosed and repaired hardware/software faults in laptops and desktops; performed OS installation and configuration.
- Gained practical experience in real-time electronics troubleshooting and system-level debugging.



PERSONAL INFO

Phone
6383075884

Email
kabi03093@gmail.com

Location
Kanchipuram, TN

LinkedIn
linkedin.com/in/kabilesh-s

GitHub
github.com/kabilesh-s

TECHNICAL SKILLS

C / C++
Arduino / ESP32
UART / SPI / I2C
PCB Design (KiCad)
Circuit Design
Digital Electronics
MATLAB / Simulink
IoT Systems
GSM / GPS Modules
Hardware Debugging
VS Code / GitHub

CERTIFICATIONS

NPTEL – Embedded Systems
IIT / SWAYAM | 2023
Smart India Hackathon
Participant | 2024

Hacker Rank
25 Modules Completed | 2023

Leet Code
Problem Solving | Ongoing

LANGUAGES

Tamil
English

ACHIEVEMENTS

IoT & Embedded Hackathons
2023 – 2024
Hardware Prototyping under
time-constrained environments